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Motion Control of Wheeled-Legged Quadrupeds on Complex Terrains via Integrated Latent Encoding

Xinyuan Shi[a], Shijun Xiao[a], Hui Ding[a], Teng Chen[a], Chao Ding[b], Guoteng Zhang[a], Xuewen Rong[a,*]

[a]*School of Control Science and Engineering, Shandong University, Jinan, 250061, China*
[b]*College of Electrical Engineering and Automation, Shandong University of Science and Technology, Qingdao, 266590, China*

Abstract

Effectively controlling wheeled-legged robots on complex terrains remains an ongoing challenge in robotics. To enhance the motion performance of wheeled-legged quadrupeds on complex terrains, this paper proposes a control method that leverages integrated latent encoding and teacher-student networks within a privileged learning framework to achieve robust adaptability. By leveraging a privileged learning framework, the robot autonomously acquires robust motion skills and adaptability on complex terrains. A regionalized curriculum learning strategy is designed to ensure the robot can move at high speeds while adapting to complex terrains. Simulation and real-world platforms using IsaacGym, Webots, and a physical robot have been established for complex terrain motion tests of wheeled-legged quadrupeds. Experiments have demonstrated the efficacy of the proposed control method in enabling wheeled-legged quadrupeds to robustly traverse complex terrains such as rugged grasslands, stairs, and slopes.

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Keywords: Wheeled-legged robots; Reinforcement learning; Hybrid motion

1. Introduction

Wheeled robots are widely adopted in flat terrains due to their high motion efficiency, strong load carrying capacity, and relatively simple motion control. However, their poor obstacle crossing ability and limited adaptability to uneven terrains significantly constrain their applicability on complex environments. In contrast, legged robots excel in terrain adaptability and obstacle negotiation, enabling robust traversal over rugged surfaces, albeit at the cost of reduced motion efficiency. To leverage the complementary strengths of wheeled and legged morphologies, researchers have proposed the development of wheeled-legged robot configurations [1, 2, 3, 4]. Boston Dynamics' Handle robot exemplifies this design paradigm by employing a two-wheeled drive system that enables rapid motion, dynamic balancing, and agile maneuvers such as jumping and spinning. Similarly, Swiss-Mile, a quadrupedal wheeled-legged robot developed by a spin-off from ETH Zurich, demonstrates impressive versatility. In parallel, Unitree Robotics has introduced

* Corresponding author. Tel.: +86-177-0531-6578.
E-mail address: rongxw@sdu.edu.cn

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B2-W, a quadrupedal wheeled-legged robot capable of executing 360° spinning jumps. DEEP Robotics has developed Mountain Cat, an all-terrain off-road robot designed to climb steps and slopes. These representative wheeled-legged robots, by integrating the strengths of both wheeled and legged motion, have shown exceptional potential across diverse application domains such as disaster response, logistics, and industrial inspection.

Due to the complexity of integrating legged and wheeled structures, wheeled-legged quadrupeds have intricate kinematic and dynamic characteristics. Traditional control methods based on precise analytical models can effectively manage the motion of wheeled-legged systems under certain conditions. However, they lack the flexibility for real-time gait transitions on complex environments. To address this challenge, this paper presents a control framework for wheeled-legged quadrupeds on complex terrains, integrating integrated latent encoding into a privileged learning–based training strategy. The main contributions are as follows:

- A privileged learning with integrated latent encoding framework is proposed, using teacher policy experience to guide student policy decisions via knowledge distillation and incorporating latent code aggregation to enhance learning efficiency. This approach significantly enhances the wheeled-legged quadrupeds' motion on complex terrains.
- A region-based curriculum learning strategy is designed to enable the robot to achieve high-speed motion and environmental adaptability across diverse terrains, and a complete deployment framework integrating virtual simulation with real-world deployment is established to validate the strategy's effectiveness and robustness.

2. Related Work

2.1. Traditional Control Methods

Model-based control excels in managing wheeled-legged system motion. By establishing precise kinematic and dynamic models, these methods can accurately describe robot behavior during various motion states. Bjelonic et al. developed a whole-body model predictive control (MPC) approach [5] that optimizes wheel and torso motions by planning joint velocities and ground reaction forces in real time, enhancing robot stability and adaptability during high-speed motions. Belli et al. proposed a trajectory generation method based on extended single rigid-body dynamics with non-holonomic constraints [6]. Zhang et al. incorporate a Stribeck-friction model identified from the robot dynamics as feedforward into a linear–quadratic regulator [7], markedly enhancing low-speed postural stability. Liu et al. present an adaptive dynamic optimal balance controller [8], integrating virtual model control with a virtual-leg PD+P support-force scheme for online gain-matrix adaptation to external disturbances.

2.2. Reinforcement Learning Methods

Reinforcement Learning (RL) is a data-driven framework where agents learn policies through environment interactions to maximize long-term rewards, making it suitable for complex and dynamic settings. RL algorithms are effective for high-dimensional nonlinear systems and optimizing motion patterns. In quadruped control, RL-based methods have shown superior performance [9], offering new solutions for wheeled-legged quadrupeds motion control. ETH used model-free RL to develop motion controllers, training policies in simulation with full-body dynamics and stochastic perturbations before transferring them to physical robots [10]. Building on this, the Zurich team proposed an integrated control system [11], combining adaptive motion, local navigation with terrain-aware perception, and path planning, creating a unified motion controller for robust and efficient motion in highly unstructured terrains.

3. Method

The overall control architecture of the system is shown in Fig. 1. A training environment is built from the robot's physical parameters and actual terrain requirements. A privileged-learning framework with aggregated latent encoding trains a network to produce the target policy, which is exported, deployed, and validated in high-fidelity simulation and on the robot.

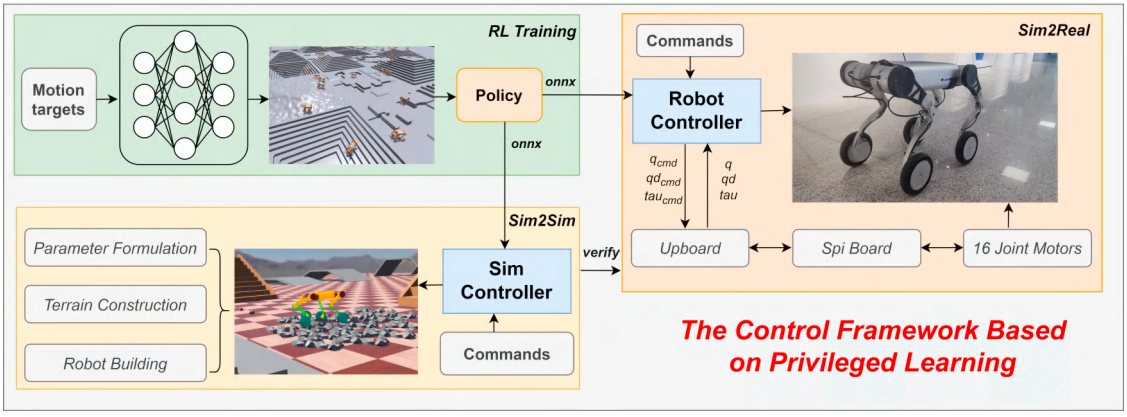


Fig. 1. Overview of the control framework

3.1. Privileged Learning with Integrated Latent Encoding

In our work we concentrate on blind gait adaptive control algorithms for wheeled-legged quadrupeds, aiming to handle dynamic balance challenges in unstructured terrains. As the robot lacks comprehensive semantic perception in unknown environments, its motion planning relies on a Partially Observable Markov Decision Process (POMDP) framework. To overcome the slow convergence of traditional reinforcement learning in latent state spaces, a knowledge distillation based privileged learning architecture [12] is introduced.

Teacher-Student Network. In this architecture, a teacher policy π_θ^T is trained with prior perception of environmental dynamics. It uses privileged information x_t like ground parameters, robot physical data, and interaction states from the simulation, along with proprioceptive information, to develop robust terrain adaptive motion. A student policy π_θ^S imitates the teacher's decision making by learning from historical proprioceptive data o_t, enabling the physical system to acquire the teacher policy's terrain adaptation capabilities despite lacking information privileged. The overall network structure is shown in Fig. 2. The forward propagation of these networks is expressed as follows:

$$l_T = E_\theta(x_t, a_{t-1}), l_S = I_\theta(o_t, a_{t-1}) \tag{1}$$

Here, E_θ denotes the teacher encoder network, and I_θ represents the student distillation network. l_T and l_S are the latent variables output by the encoding and distillation networks. At the beginning of training, the policy relies entirely on l_T, derived from the privileged observations via the teacher encoder, for updates. Subsequently, using the iteration counter as a threshold, l_S, the output of the student distillation network based on the robot's proprioceptive observations, is progressively incorporated into the optimization objective. By employing a mean squared error (MSE) loss function, l_S is gradually aligned in distribution with l_T, thereby achieving the progressive aggregation of the two latent spaces and ensuring the continuity of the policy input distribution throughout the training trajectory.This process continues until the student policy performs well on unknown terrains, replicating the teacher's behavior. Meanwhile, latent codes from this process are used by the critic network for policy evaluation and the actor network for action selection. The output of the actor network A_θ is given by:

$$a_t = \pi_\theta^T(o_t, x_t) = A_\theta(o_t, l_T) \tag{2}$$

Empirical Normalization. To enhance training stability and data efficiency, we introduce empirical normalization. During training, we continuously receive new data batches and compute their mean and variance in real-time. Input data is then standardized during the forward pass to stabilize its distribution.

Domain Randomizations. To improve policy generalization and optimize sim-to-real deployment, training introduces diverse random perturbations for policy learning under various conditions. Specific data is shown in Table 1.

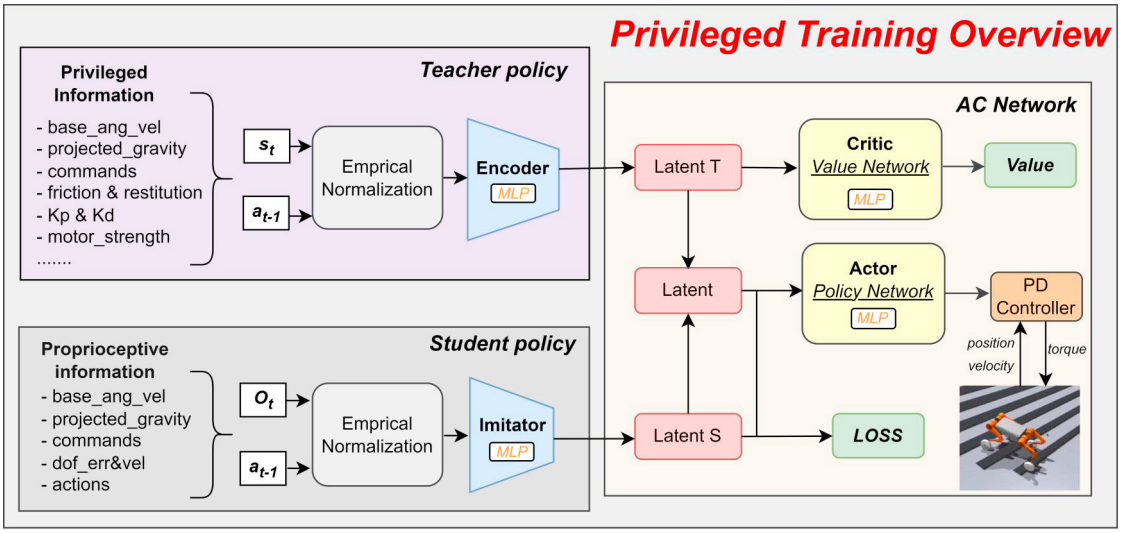


Fig. 2. Overview of the privileged learning method. During the early training phase, the policy network directly utilizes the latent variable Latent T generated by the teacher encoder, i.e., Latent=Latent T. In the later training phase, the policy network progressively transitions to using the latent variable Latent S produced by the imitator, i.e., Latent=Latent S.

Table 1. Domain Randomization Parameters.

Parameters	Range[Min, Max]	Unit
Ground Friction	[0, 2.0]	-
Payload Mass	[-1.5, 1.5]+robot mass	Kg
Motor Strength	[0.9, 1.1]×motor torque	Nm
Joint K_p	[0.8, 1.2]×30	-
Joint K_d	[0.8, 1.2]×0.75	-
Action Delay	[0, 4]	step

3.2. Regionalized Curriculum Learning

In wheeled-legged quadrupeds training, curriculum learning enhances model generalization and adaptability [13]. Training is divided into phases with progressive terrain complexity and motion commands, enabling skill acquisition from simple to complex. Given the need for both high-speed movement and terrain adaptability, a regional curriculum strategy is designed. Robots train in two terrain regions: a flat area for high-speed training and a complex area with obstacles like stairs and slopes, organized by difficulty. Robots start in lower-difficulty terrains and advance as they adapt. When the robot's score for tracking linear velocity commands exceeds 0.8 of the maximum possible score, it is moved to a more difficult terrain.The speed commands for the flat and complex regions are shown in Table 2.

Table 2. Velocity Commands for Different Regions.

Region Type	Forward Velocity Range (m/s)	Lateral Velocity Range (m/s)	Rotational Angular Velocity Range (rad/s)
Flat Area	[-1, 4]	[-1, 1]	[-3.14, 3.14]
Complex Area	[-0.5, 1.5]	[-0.75, 0.75]	[-3.14, 3.14]

3.3. Reward Terms Design

To improve the gait and stability of wheeled-legged quadrupeds, it incorporates rewards for foot suspension time, body height, and posture, adjusted by terrain level. It also constrains joint torque, speed, and position, and includes

interaction-related rewards to minimize collision impacts and prevent wheel slippage or improper leg lifting. The specific reward functions are shown in Table 3.

Table 3. Reward Functions.

No.	Reward Function Name	Calculation Formula	Weight
1	liner velocity tracking	$\exp(-\|\|v_{xy} - v_{cmd}\|\|/0.25)$	2.0
2	base height	$10 \cdot (h_{base} - h_{target})^2 / exp(L_{terrain})$	0.5
3	orientation	$2 \cdot sum(g_{xy})^2 / exp(L_{terrain})$	-2.0
4	linear velocity z	$(v_z)^2$	-2.0
5	dof torques	$\sum \tau^2$	-1e-6
6	dof velocity limits	$\sum(\dot{q} - \dot{q}_{lim})^2$	-0.5
7	feet contact forces	$\sum(\|\|F_{contact}\|\| - F_{max})$	-0.01
8	stumble	1 if $F_{xy} > 5 \cdot F_z$ else 0	-0.5
9	action rate	$\sum(a_{last} - a)^2$	-0.02
10	high speed	$\exp(-\sum(q - q_{def})^2) + 0.004 \cdot \sum \|\dot{q}_{wheel}\|$ if $\|\|v_{cmd}\|\| > 0.5$	0.1

4. Experiments

To validate the effectiveness of the proposed method, experiments were conducted in both simulation and real robot platforms.

4.1. Training

In this study, we used a single NVIDIA RTX 3080 GPU with IsaacGym as the simulation platform to train 4,096 simulated robots. To achieve good training results, about 10,000 iterations were needed, taking roughly 7 hours. Each sampling step was performed at 100 Hz, while joint position and velocity commands were sent at 500 Hz. To evaluate each component's impact on strategy performance in the training framework, we adopted a systematic experimental approach. We added different components one by one to the basic training framework and compared their effects on the reward value. The experimental method is as follows:

AC: This method includes only the basic policy and value networks, forming the simplest training framework.

AC+RC: Based on the baseline method, this approach adds regionalized curriculum learning to enhance the model's adaptability to different terrain conditions.

TS: This framework introduces encoder, policy, distillation, and value networks, creating a teacher-student training mode. It aims to improve strategy learning efficiency through knowledge distillation.

The proposed method(TS+RC): This combines the teacher-student training framework with regionalized curriculum learning, leveraging privileged information to assist training while optimizing task adaptability.

Except for the network architecture and curriculum settings, all methods use the same reward function coefficients and algorithm hyperparameters. Each method is trained five times with different random seeds, and the average rewards are shown in the Fig. 3. The results indicate that the ranking of the rewards obtained by the four methods is as follows: the highest reward is achieved by the TS+RC method, followed by the AC+RC method, then the TS method, and finally the AC method. This indicates that teacher-student strategies with privileged information have a better decision making ability.

4.2. Sim2Sim

In reinforcement learning training deployment, Sim2Sim methods are widely used [14]. Testing trained strategies in simulation first avoids high real world costs and safety risks, especially in scenarios with complex operations or valuable equipment. Complex terrains can be easily constructed in simulation for more convenient generalization testing. This paper builds a simulation environment in Webots, where the simulated robot controller runs at 500 Hz and the strategy updates at 100 Hz, matching the physical prototype. As shown in the Fig. 4, motion tests include climbing 17 cm stairs, crossing single beam bridges, overcoming 0.35 m platforms, traversing 30 degree slopes, and high speed movement (up to 4 m/s).

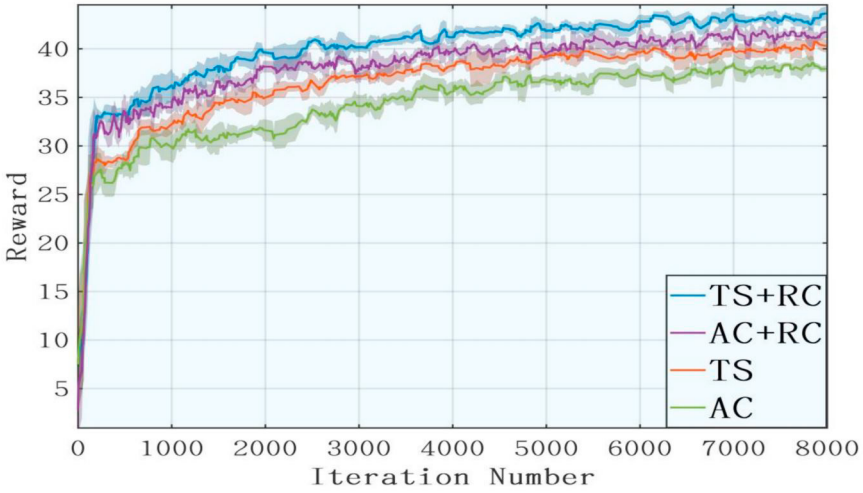


Fig. 3. Average reward curve

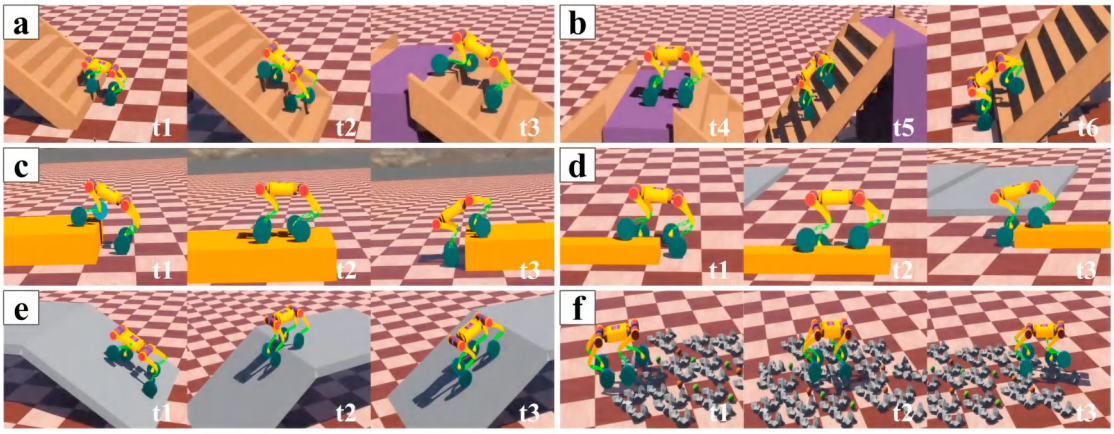


Fig. 4. motion experiments in Webots.(a)(b) 17 cm stairs; (c) 0.35 m platforms; (d)single beam bridges; (e)30 degree slopes; (f)rocky terrain

4.3. Sim2Real

The SDUQuad48w robot, standing at 0.37 m and weighing 22 kg, is equipped with four brushless DC motors on each leg for the roll hip, pitch hip, pitch knee, and active wheel joints. The motion controller runs at 500 Hz with strategy updates at 100 Hz. The trained strategy network is exported and optimized using ONNX Runtime, then reloaded into a finite state machine on the UP Board. To test the robot's motion robustness, experiments were conducted, which includes moving on rough grass, cobblestone road, sandy land, 12-cm steps, and slopes. As shown in the Fig. 5, the robot could robustly navigate these terrains, lifting its legs to maintain balance when facing obstacles and driving wheels directly on unobstructed paths to save energy. The prototype's motion closely matched simulations, confirming the training framework's strong robustness and good generalization ability.

5. Conclusion

This paper introduces a privileged learning-based control method to enhance the motion capability of wheeled-legged quadrupeds on complex terrains. The method integrates latent encoding into teacher-student networks and employs regionalized curriculum learning to improve the robot's adaptability and motion performance. A reward function is designed to optimize wheel speed, action frequency, and energy use. Experiments in IsaacGym, Webots,

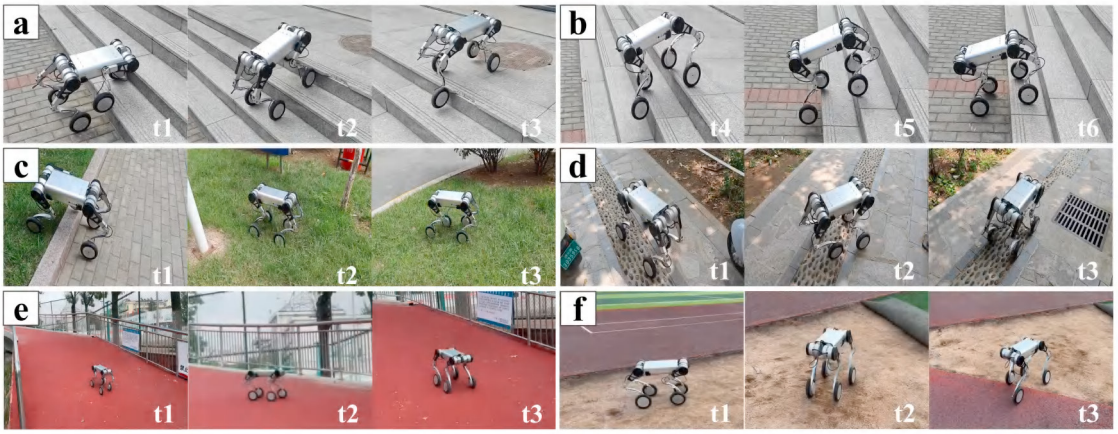


Fig. 5. motion experiments in real world.(a)(b)12cm stairs; (c)rugged grassland; (d)cobblestone road; (e)slope; (f)sandy land

and real-world platforms validate the method's effectiveness. Future research will investigate more efficient Sim2Real transfer methods to minimize simulation-reality gaps and improve strategy performance in physical robots.

Acknowledgements

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